

SkillClaw: Collective Skill Evolution for Multi- User LLM Agent Ecosystems

Enabling LLM agent skills to continuously evolve by aggregating cross-user interaction trajectories and applying agentic, evidence-driven skill updates

arxiv.org/abs/2604.08377

Agent Skills Remain Static After Deployment

Users rediscover solutions independently — knowledge never accumulates at the system level

CURRENT PARADIGM

Static Skill Libraries

- Skills manually installed from a centralized hub
- Solutions discovered in one session rarely persist
- Same failure patterns recur across users over time
- Memory-based methods stay tied to specific instances

NEEDED PARADIGM

Evolving Skill Ecosystems

- Skills that improve through runtime interaction
- Cross-user knowledge accumulation
- Failure patterns diagnosed and corrected automatically
- Experience compressed into reusable, improving skills

Each user is forced to rediscover solutions independently, preventing knowledge from accumulating at the system level.

Closed-Loop Pipeline: Multi-User Interaction to Shared Skill Evolution



Agentic Evolver: Evidence, Attribution, Evolution in Three Stages

STAGE 1

Evidence

Analyze aggregated session trajectories grouped by referenced skill. Identify behavioral patterns — which conditions the skill handles well and where it breaks down.

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STAGE 2

Attribution

Diagnose root causes behind observed failures and successes. Determine whether issues stem from the skill itself or from environmental context.

→

STAGE 3

Evolution

Propose skill mutations — refining existing skills or creating new ones. Candidate skills are verified before merging into the shared repository.

The evolution process is driven by an autonomous agent performing **open-ended reasoning** over interaction evidence — not predefined update rules.



Cross-User Trajectory Aggregation Exposes Skill Behavioral Boundaries

- OpenClaw – success under standard inputs
- CoPaw – failure on edge-case formatting
- IronClaw – success with extended parameters
- PicoClaw – failure on auth-restricted paths
- ZeroClaw – success on clean environments
- NanoClaw – failure on large file inputs
- NemoClaw – success with fallback paths

Complementary Signals

Different users exercising the same skill under diverse contexts produce complementary views of that skill's behavioral boundary — revealing both conditions under which it works and those under which it breaks.

Single-User Insufficiency

A single user rarely generates enough signal to separate a generalizable improvement from an idiosyncratic fix. Cross-user aggregation resolves this ambiguity.

- OpenClaw
- CoPaw
- IronClaw
- PicoClaw
- ZeroClaw
- NanoClaw
- NemoClaw



Case Study: Multimodal Creative Pipeline

1 of 6 accepted

6 days of evolution attempts — conservative verification prevents regressions

Day	Candidate Skill	Outcome	Rationale
Day 1	validate-tmp-workspace-inputs	Accept	Input validation upgrade — promoted to new best pool
Day 2	multimodal-input-validation-and-setup	Reject	No improvement over current best pool
Day 3	multimodal-creative-task-pipeline	Reject	Full pipeline replacement insufficient
Day 4	multimodal-creative-task-pipeline (impr.)	Reject	Added classification & generation — still insufficient
Day 5	pipeline (impr.) + validate-required-input-files	Reject	Dual candidate — fail-fast validation not enough
Day 6	multimodal-creative-task-pipeline (cand.)	Reject	Extended PDF/poster paths — not admitted to next cycle



Case Study: Git Push Auth-Fallback

3 of 6 accepted

Iterative refinement succeeds before reaching a plateau

Day	Candidate Skill	Outcome	Rationale
Day 1	git-push-with-auth-fallback	Accept	Safe fallback instead of blocking on push failure
Day 2	git-push-with-auth-fallback	Accept	Unified filenames, reduced inconsistency
Day 3	auth-fallback + git-clone-to-directory	Accept	Alternative paths & secrets audit; fixed subshell pitfalls
Day 4	(same-pool retest)	Accept	Validator confirmed current pool as best — continue
Day 5	git-push-with-auth-fallback	Reject	Added push-hang-as-auth-failure — no improvement
Day 6	git-push-with-auth-fallback	Reject	Identity config & filename consistency — no improvement



Three Properties That Distinguish SkillClaw

PROPERTY 1

01

Collective Evolution

Knowledge from individual interactions contributes to a shared, continuously improving skill ecosystem. Every session makes the system better for all users.

PROPERTY 2

02

Fully Automatic

Skill evolution is driven by runtime interaction without manual curation or explicit user intervention. No human in the loop needed.

PROPERTY 3

03

Agentic Evolution Paradigm

Skill updates are produced through open-ended reasoning rather than predefined update rules. The evolver adapts its strategy to the evidence.

Implication: SkillClaw demonstrates that continuously improving agent systems are achievable when individual interactions feed a shared, agentic evolution loop — moving from static tooling to self-improving ecosystems.